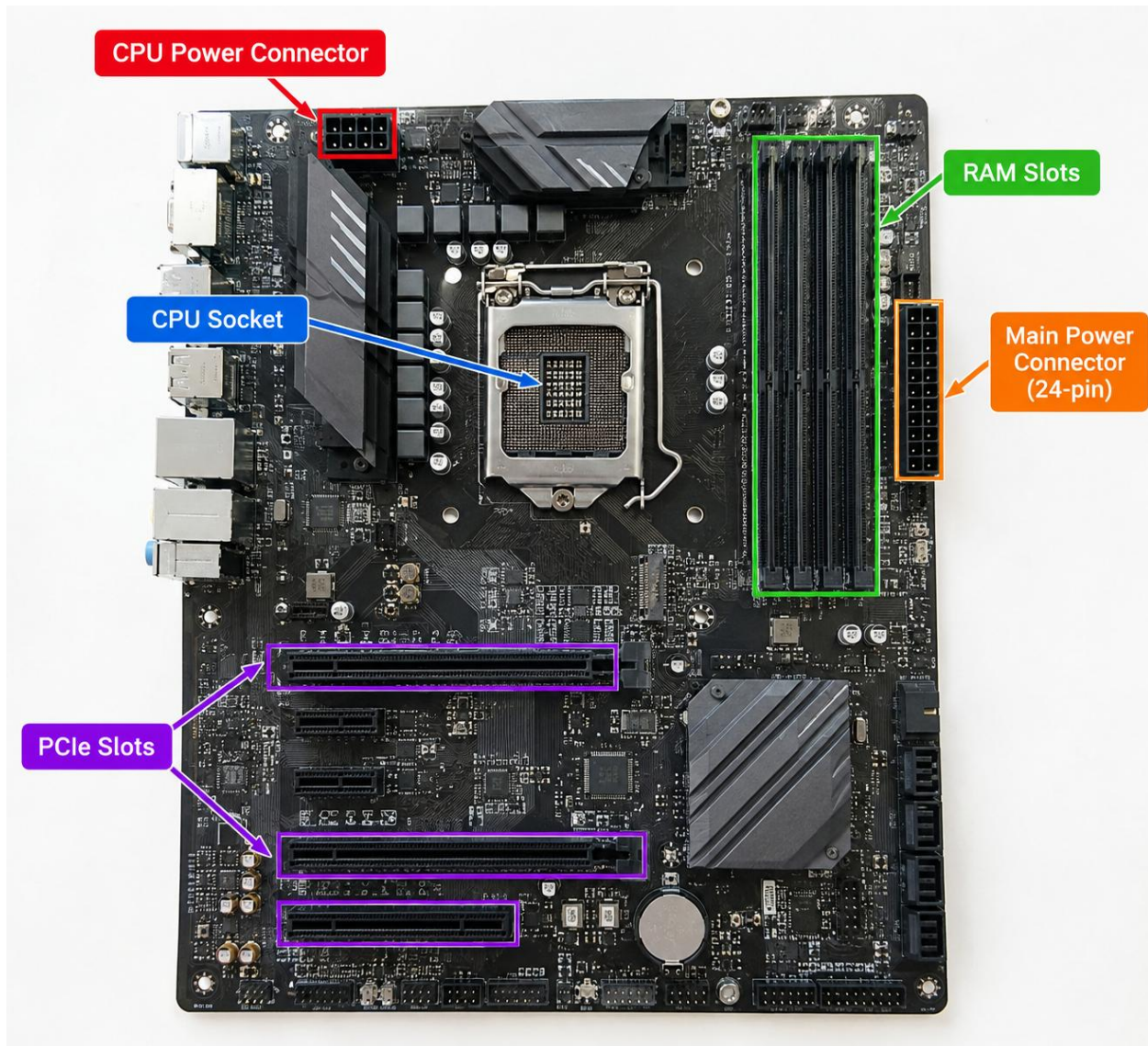


Motherboard and CPU

1. Take a clear photo of the motherboard inside your own or a friend's laptop/desktop (or find a labeled image online), and label the CPU socket, RAM slots, PCIe slots, and power connectors using any image editing tool.

->



Here is a **ready-to-use labeled motherboard image** for your assignment. It clearly labels the **CPU Socket, RAM Slots, PCIe Slots, and Power Connectors**. These are the same major motherboard areas described in standard motherboard layouts.

You can use this image as your assignment reference or insert it into an image editor and add your name/class details.

2. Pick any three popular CPUs used in gaming laptops or desktops (for example, Intel Core i5-12400F, AMD Ryzen 5 5600, Intel Core i7-13700K). Research and create a table comparing their clock speed (GHz), number of cores/threads, cache size (L2/L3), TDP (W), and current price in India.

->Here is a clear comparison table you can use for your assignment. The prices are **current Indian online prices found during September 2026** and can change with stock and discounts.

CPU	Clock Speed (Base / Boost)	Cores / Threads	L2 Cache	L3 Cache	TDP	Current Price in India
Intel Core i5-12400F	2.5 / 4.4 GHz	6 / 12	7.5 MB	18 MB	65 W	₹15,500
AMD Ryzen 5 5600	3.5 / 4.4 GHz	6 / 12	3 MB	32 MB	65 W	₹13,131
Intel Core i7-13700K	3.4 / 5.4 GHz*	16 / 24	24 MB	30 MB	125 W	₹33,985

Sources: The i5-12400F specifications and price are listed by PrimeABGB. The Ryzen 5 5600 specifications and current listed price are from PrimeABGB. The i7-13700K specifications and listed Indian price are from PrimeABGB, while Intel confirms its 16-core/24-thread, 24 MB L2, 30 MB cache and 125 W base-power specifications.

*The i7-13700K has **3.4 GHz P-core base frequency**; its maximum turbo frequency is **5.4 GHz**.

Short conclusion

- **Core i5-12400F:** 6 cores and 12 threads, with a 65 W base power, making it a commonly used mid-range desktop CPU.
- **Ryzen 5 5600:** 6 cores and 12 threads, 32 MB L3 cache and 65 W TDP.
- **Core i7-13700K:** 16 cores and 24 threads with a 5.4 GHz maximum turbo frequency and higher 125 W base power.

Note: CPU prices in India vary by retailer, availability, GST, and discounts, so the prices above should be treated as a snapshot rather than fixed prices.

3. For each CPU in your comparison table, identify at least one compatible motherboard model (mention brand and model, e.g., ASUS PRIME B660M-A for Intel 12th Gen) and specify the socket type and form factor (ATX, mATX, ITX).

->Here is a suitable motherboard for each CPU. I checked the manufacturers' specifications for socket compatibility and form factor.

CPU	Compatible Motherboard	Brand	Socket Type	Form Factor
Intel Core i5-12400F	PRIME B660M-A D4	ASUS	LGA1700	mATX
AMD Ryzen 5 5600	PRIME B550M-A	ASUS	AM4	mATX
Intel Core i7-13700K	PRIME Z790-P WIFI	ASUS	LGA1700	ATX

Details

1. **Intel Core i5-12400F → ASUS PRIME B660M-A D4**

This motherboard uses the **LGA1700 socket**, supports 12th-generation Intel Core processors, and has an **mATX** form factor. ([ASUS Global](#))

2. **AMD Ryzen 5 5600 → ASUS PRIME B550M-A**

ASUS lists the **Ryzen 5 5600** in the CPU support list for the PRIME B550M-A. It is an **AM4** motherboard in the **mATX** class. ([ASUS Global](#))

3. **Intel Core i7-13700K → ASUS PRIME Z790-P WIFI**

The PRIME Z790-P WIFI uses **LGA1700** and supports 13th-generation Intel Core processors. ASUS specifies its form factor as **ATX**. ([ASUS Global](#))

Assignment-ready conclusion: These examples show that motherboard compatibility depends primarily on the CPU's **socket type and supported CPU generation**. The i5-12400F and i7-13700K both use LGA1700, while the Ryzen 5 5600 uses AM4.

4. Visit the official website of any one CPU you compared (Intel or AMD), and find the maximum supported RAM speed and PCIe version for that CPU. Write these details along with how they affect real-world performance in gaming or multitasking.
Hint: Look for 'Memory Types' and 'PCI Express Revision' in the CPU specs page.

->For this comparison, I'll use the **Intel Core i5-12400F** and its official Intel specifications.

Intel Core i5-12400F – RAM and PCIe Specifications

According to the official Intel specifications, the **Intel Core i5-12400F** supports up to **DDR5-4800 MT/s** or **DDR4-3200 MT/s** memory. It also supports **PCI Express 5.0 and 4.0**, with up to 20 PCIe lanes. ([Intel](#))

Specification	Intel Core i5-12400F
Maximum supported RAM speed	DDR5-4800 MT/s
DDR4 support	Up to DDR4-3200 MT/s
PCIe version	PCIe 5.0 and PCIe 4.0
Maximum PCIe lanes	20

Effect on Real-World Performance

RAM: Faster RAM can improve memory bandwidth and responsiveness, which can help gaming performance in some CPU-limited situations and make multitasking smoother. However, the actual improvement depends on the game, applications, motherboard, and memory configuration.

PCIe: PCIe 5.0 provides a high-speed connection for compatible expansion devices such as graphics cards and NVMe SSDs. In everyday gaming, the difference is not always large because many current GPUs and SSDs do not require the full bandwidth of PCIe 5.0. It is more useful for providing additional bandwidth for future or high-speed hardware.

Conclusion: The i5-12400F's support for DDR5-4800 and PCIe 5.0 gives it good memory and expansion capabilities. However, real-world gaming performance also depends heavily on the GPU, game settings, RAM configuration, and other components.

Official source: Intel's Core i5-12400F specification page lists the memory types and PCI Express revisions directly. ([Intel](#))

If you want, I can also make this into a **short 5–6 line assignment answer**.